The Reggae Shark Protocol (v1.3.0 — Cross-Program 3-Track Upgrade): 41 Lean 4 Theorems, 200-Bit Interval-Certified Patankar–Euler & WP19 Bridges, Multi-Principal Subprocess Ed25519 Signing, and Bandlimited LRSC Compression

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Abstract

The Reggae Shark Protocol (v1.3.0) is a verifiable execution kernel and multi-cloud AI gateway that binds agent prompt dispatches to four verification layers and three cross-program mathematical/security upgrades: (1) an append-only, SHA-256 hash-chained and RFC 8032 Ed25519-signed **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger enforcing `intentseal-vrk-claim-v2`, prompt/response commitments, on-disk evidence anchor checks, an N-way associative Merkle-DAG join-semilattice, and **Track B multi-principal OS subprocess key isolation** (`IsolatedSignerSubprocess`); (1B) a thread-synchronized **Request-Lifecycle Typestate and Worker-Epoch Isolation** state machine (`TypestateLifecycle.lean`, expanded to 15 zero-`sorry` theorems); (2) a bounded-LRU per-tenant **4D Soft-Wall ODE Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 4D boundary theorems (`ExistenceVerification.lean`), 9 3D soft-wall theorems (`BoundaryVerification.lean`), and **Track A 200-bit outward-rounded `mpmath.iv` Patankar–Euler interval enclosures** plus **Track C 120-segment piecewise Duhamel contraction certificates** for Navier–Stokes `WP19`; and (3) a **Bandlimited LRSC** operator ($M = 99, K = 11$) with a bounded-domain local sensitivity bound.

1 Four-Layer Verification Architecture & Provenance (41 Lean 4 Theorems)

Every request submitted to `MultiProviderGateway.process_request` or the local loopback HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant
Layer 1	`shark_protocol/kernel.py`	**Ed25519-Signed VRK v0.3.1 Chain, Multi-Principal IPC & Merkle-DAG Join:** Append-only SHA-256 chain anchored at trust root `f010a99d...05a0d0f` (10.5281/zenodo.23018459), committing to `prompt_sha256`, `response_sha256`, on-disk `evidence_root`, Ed25519 signatures (with optional `IsolatedSignerSubprocess`), and verified `parent_hashes`.
Layer 1B	`lean/TypestateLifecycle.lean`	**Typestate, Epoch & Cross-Program Bridge Suite (15 Theorems) + RLock:** 6-state request automaton (`UNINIT` $\to \cdots \to$ `ZEROED_FREED`) with absorbing `ZEROED_FREED`, inductive trace safety (`runTrace`), and Theorems 13–15 for Tracks A, B, and C.
Layer 2	`lean/ExistenceVerification.lean` `lean/BoundaryVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$, 26 Theorems) & 200-Bit Interval Bridge:** Adaptive RK4, float64 Patankar–Euler, and 200-bit `mpmath.iv` interval enclosure (`step_ode_patankar_mp_interval`) inside $\mathcal{B}_4 = [0,10] \times [0,100] \times [0, \dot{I}_{\max,4D}] \times [0,1]$.
Layer 3	`LRSCCompressor` (`kernel.py`)	**Bandlimited LRSC Odd-Ring Projection ($11/99 = 1/9$):** Projects context onto the $M = 99$ odd-ring filter V_{ring} ($\sigma_{\min}(S_{11}) = 0.181646 > 0$, 10.17605/OSF.IO/NM5BW), evaluates a bounded-domain local sensitivity proxy [6], and binds distinct compressed/uncompressed SHA-256 digests.
Layer 4	`shark_protocol/gateway.py`	**Fail-Closed Multi-Tenant Gateway & ThreadingHTTPServer:** Enforces waking basal drive $\mathcal{P}_0 \geq 1.0 \gg 10/99$, bounded tenant LRU (`max_tenants=128`), header-only API keys (`x-goog-api-key`), HTTP 413 payload caps, and fail-closed live dispatch.

Table 1: Four-layer verification architecture of The Reggae Shark Protocol (v1.3.0).

Canonical Lean 4 Module Provenance (41 Zero-`sorry` Theorems): The 24-file release archive contains 23 hash-verified files plus `MANIFEST_SHA256.txt`. Specifically, `lean/BoundaryVerification.lean` (`SHA-256: 49229b61...e417`, 9 theorems) is byte-for-byte identical to Physics v1.5 Appendix A (10.5281/zenodo.23534449), and `lean/ExistenceVerification.lean` (`SHA-256: eec94868...bcae`, 17 theorems) is byte-for-byte identical to *The Theory of Bounded Existence* v1.0.0 (10.5281/zenodo.23608244). Together with the 15 theorems in `lean/TypestateLifecycle.lean` (each verified with `#print axioms`), the package compiles **41 zero-`sorry` Lean 4 theorems**.

2 What Is and Is Not Modelled (Epistemic Claim Firewall)

Epistemic Standard (`Build. Attack. Falsify. Preserve.`): Every theorem and runtime guard in this repository is bounded by explicit domain assumptions and `scope.does_not_imply` declarations.

- **Application-Layer Typestate & Subprocess Signer vs. OS Ring-0:** `RequestSlot` and `WorkerEpochGuard` enforce typestate invariants in Python, while `IsolatedSignerSubprocess` isolates the Ed25519 private key in a separate OS child process (`subprocess.Popen`) across a narrow pipe IPC boundary. They are *not* an OS ring-0 kernel module or hardware HSM.

- **Real Arithmetic in Lean vs. 200-Bit `mpmath.iv` & IEEE-754 Runtime Enclosures:** Lean 4 Theorems 6, 7, 9, 10, 13, and 15 are proved over $\mathbb{R}$; `step_ode_patankar_mp_interval` bridges discrete steps to $\mathbb{R}$ via 200-bit outward-rounded `mpmath.iv` intervals (`INTERVAL_CERTIFIED_RESULT`).
- **Piecewise Duhamel Finite-Cutoff Tightening vs. Continuum Navier–Stokes:** `certify_wp19_piecewise_duhamel_bridge` certifies 120-segment Duhamel contraction for the fixed finite Galerkin chain (10.5281/zenodo.23034545); cutoff-uniform continuum 3D Navier–Stokes regularity remains `OPEN`.

3 Layer 2 Equations, Five Core Guards & Three Cross-Program Upgrades (34/34 Tests)

The Layer 2 governor evolves $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall system [3]:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \qquad \frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}\tanh\!\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{1}$$

$$\frac{d\dot{I}}{dt} = e\tanh\!\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma,\Psi) - g\dot{I}, \qquad \frac{d\Psi}{dt} = \kappa_{\text{meta}}R_{\text{self}}(x)\,(1-\Psi) - \mu_\Psi\Psi, \tag{2}$$

where $\mathcal{P}_{\text{eff}}(\gamma,\Psi) = \mathcal{P}_0 + \lambda_{\text{sim}}r_{\text{LRSC}}\gamma + \lambda_{\text{lucid}}\Psi$, $R_{\text{self}}(x) = \text{fb}(x)^2/(\text{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$, and $\text{fb}(x) = e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$.

1. **Guard 1, Probe A & Track A (200-Bit `mpmath.iv` Patankar–Euler Real-to-Float Bridge):** Updates each coordinate via

$$x_{k+1} = \frac{x_k + \Delta t\,P(x_k)\,x_{\max}}{1 + \Delta t\,(P(x_k) + D(x_k))}, \qquad P(x_k) \geq 0,\ D(x_k) \geq 0,\ 1 + \Delta t(P+D) \geq 1, \tag{3}$$

   paired with Python float64 floor `FLOAT64_NORMAL_MIN` $= 2.225 \times 10^{-308}$ and `step_ode_patankar_mp_interval` (200-bit outward-rounded `mpmath.iv` intervals of width $< 10^{-45}$ agreeing with float64 within $< 10^{-12}$, Theorem 13 `patankar_interval_rounding_in_box`, `test_15`, `test_21`, `test_31`).
2. **Guard 2 & Probe B (Pointwise Decay Bound, Bounded Tenant LRU & Waking Drive Floor):** Theorem `psi_decay_floor_bound` proves $d\Psi/dt \geq -\mu_\Psi\Psi$ (yielding $\Psi(t+\Delta t) \geq \Psi(t)e^{-\mu_\Psi\Delta t}$ via Grönwall), while `MultiProviderGateway` enforces `max_tenants=128` and $\mathcal{P}_0 \geq 1.0 \gg 10/99$ (`test_16`, `test_22`, `test_29`).
3. **Guard 3 & Probe C (Prompt-Committed Bandlimited LRSC Witness & Local Sensitivity Proxy):** `LRSCCompressor.compress` injects a slot-0 anchor to eliminate scalar-multiple collisions (`test_27`), projects onto V_{ring} ($\sigma_{\min}(S_{11}) = 0.18164624 > 0$ [4]), bounds $L_{\text{local}}\eta_\perp(\tilde{x})$ on $\|\tilde{x}\|_2 \leq 1$ [6] (Theorem 10), and seals `prompt_sha256` and `response_sha256` (`test_17`, `test_23`, `test_27`).
4. **Guard 4 & Probe D (Inductive Typestate Trace Safety, Absorbing `ZEROED_FREED` & RLock Order):** Theorems 1–5 prove over arbitrary event lists `evs : List LifecycleEvent` that `ZeroedFreed` is strictly absorbing and `{AbortedPoisoned, ZeroedFreed}` never reaches `SealedActive` or `ShutdownClean`, with Python `guard._lock` $\rightarrow$ `slot._lock` (`test_18`, `test_24`, `test_30`).
5. **Guard 5, Probe E & Track B (Ed25519 Signing, Multi-Principal Subprocess Isolation & Merkle-DAG):** Every claim is signed with Ed25519 (`test_28`), `IsolatedSignerSubprocess` isolates the private key in a child OS process across a narrow IPC pipe (Theorem 14 `principal_key_isolation_noninterference`, `test_32`), `verify_chain_list` rejects forged multi-parent tails (`test_26`), and `merge_divergent_ledgers` satisfies $(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$ (Theorem 12, `test_19`, `test_25`).
6. **Track C (120-Segment Piecewise Duhamel Contraction Bridge for Navier–Stokes `WP19`):** Theorem 15 (`duhamel_piecewise_contraction_bound`) and `certify_wp19_piecewise_duhamel_bridge` prove in 200-bit `mpmath.iv` arithmetic across the 120 segments (10.5281/zenodo.23034545) that piecewise dissipative propagation $E_{k+1} = e^{\mu_{\text{eff}}\Delta t}E_k + \delta_k$ suppresses the 1.6364×10^{14} whole-path Grönwall remainder below the signed C_{500} lower bound 5.7586×10^9 whenever $\mu_{\text{eff}} < \mu_{\text{crit}} \approx 22.95\,\text{s}^{-1}$ (`test_33`, `test_34`).

References

[1] P. Upadhyay, *IntentSeal Verifiable Research Kernel (VRK): Final Release v0.3.1*, Zenodo, 2026. doi:10.5281/zenodo.23018459.
[2] P. Upadhyay, *Appendix A v1.5 — Global Dissipative Completion and Soft-Wall Limit-Cycle Certification*, Zenodo, 2026. doi:10.5281/zenodo.23534449.
[3] P. Upadhyay, *The Theory of Bounded Existence: A 4D Closed-Loop Dynamical Model (v1.0.0)*, Zenodo, 2026. doi:10.5281/zenodo.23608244.
[4] P. Upadhyay, *LRSC Odd-Ring Spectral Rank Theorem (v1.2) & Exact M = 99 Certificate*, OSF, 2026. doi:10.17605/OSF.IO/NM5BW, doi:10.17605/OSF.IO/JAK5X.
[5] P. Upadhyay, *Navier–Stokes Bridge Audit: Finite Fourier Diagnostics (n13-same-datum-v1)*, Zenodo, 2026. doi:10.5281/zenodo.23034545.
[6] H. Kim, G. Papamakarios, and A. Mnih, *The Lipschitz Constant of Self-Attention*, Proc. ICML, PMLR 139:5562–5571, 2021.